Funny Story: The Effects of Background Laughter on Humor Processing in the Brain

MOTIVATION

Does hearing laughter make things funnier?

From a behavioral perspective, we know the sound of laughter alone is a sufficient stimulus for laughs and smiles (Provine 1992). Many popular shows have used laugh tracks, we're more likely to find amusement in something when we see others laughing along to it. But is this actually reflected in the brain? Does the mere sound of laughter heighten our neural response to humor?

BACKGROUND

A primer on fMRI

01
BOLD signal

fMRI measures the contrast between oxygenated and deoxygenated hemoglobin. Active neurons trigger a local influx of oxygenated blood — a proxy for neural activity.

02
Voxels & TRs

Brain volume is discretized into voxels (≈2–3 mm edges, ~32k neurons each). Signal is sampled every TR — 1.5 s in this dataset. Data: (x, y, z, T). Also available as GIFTI (v, T).

03
Hemodynamic lag

BOLD peaks ~5–6 s after neural activity. All GLM analyses must convolve the stimulus regressor with a hemodynamic response function (HRF).

BACKGROUND

Humor in the brain: a three-part process

Chan et al.'s tri-component framework decomposes humor processing into three stages

COMPREHENSION
the cognition
STG - IFG
MTG - TPJ

APPRECIATION
the affect
STS - vmPFC
ACC - VS/NAcc

EXPRESSION
the physical response
SMA - M1

The hypothesis: background laughter amplifies activity in comprehension and appreciation regions beyond just driving physical laughter

DATA AND PREPROCESSING

The Narratives dataset: 'Pie Man'

45 subjects randomly selected per group

23 shared punchlines

1.5 seconds per TR (sampling rate)

~80k vertices (GIFTI surface)

Live recording (The Moth)
Recorded in front of a live storytelling audience.
Volume (NIFTI): 65 × 77 × 49 × 300 TRs.
Rich audience laughter after most punchlines.
Up-sampled to match PNI resolution.

PNI Isolated Recording
Same speaker, same story, recorded without an audience at the Princeton Neuroscience Institute.
Volume (NIFTI): 78 × 93 × 78 × 294 TRs.

DATA AND PREPROCESSING

Punchline Annotation and Alignment

I defined a punchline as *the inciting incident for expected mirth*. This inflection point was the proxy for the cognitive onset of humor comprehension and appreciation.

01 **Transcribe audio**: WhisperX (large-v3) → JSON with per-word timestamps

02 **Detect laughter**: YAMNet audio-event classifier + manual verification.

03 **Punchline labeling:** Onset = unexpected word; offset = speaker's natural pause.

04 **Align across conditions:** shared_id matches the 23 punchlines in both versions.

EXPERIMENT I

Two-Level General Linear Model (GLM)

$$y_i = \beta \cdot (X_i * h) + C + \varepsilon_i$$

y_i BOLD value ∗ X_i punchline indicator ∗ h HRF convolution ∗ C confounds (pre-filtered) ∗ ε_i noise (AR(1) autocorrelated)

First Level Model: Fit β at every vertex for each subject to estimate the correlation of that vertex's activity with the punchline events

$$C_i = X_G \cdot [\beta_i^1, \beta_i^2] + \eta$$

C_i participant z-score vector ∗ X_G design matrix ∗ β_i population means ∗ η mixed effects errors

Second Level Model: two-sample t-test at each vertex comparing live vs. isolated z-scores, tests whether β_live − β_studio ≠ 0

KEY RESULT

Significantly greater punchline-vs-baseline contrast observed in humor-associated regions in background laughter condition

Note on Multiple Comparisons. At p < .05 per vertex, 80k tests ⇒ ~4,000 false positives by chance. To account for this, I used FWER correction with permutation testing, the gold standard for fMRI analyses.

Differing regions:

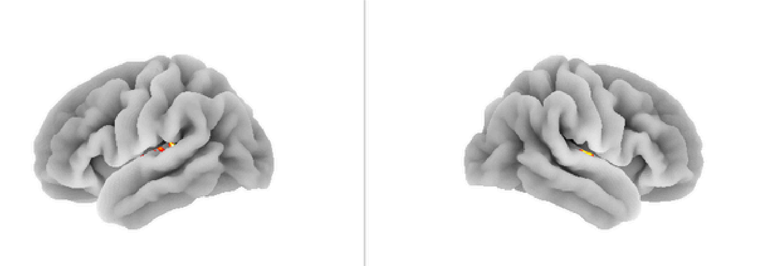
Bilateral STG *comprehension*	L: 163 vertices R: 87 vertices
Bilateral Heschl's Gyri *comprehension/auditory*	L: 78 vertices R: 43 vertices
Tranverse Temporal Sulcus *STG-adjacent*	L: 75 vertices R: 41 vertices
Precentral Gyrus *primary motor cortex*	L: 10 vertices

Interpretation: hearing background laughter results in robust amplification of activity in humor-associated regions (with probability of even one false positive thresholded at 5%).

RELIABILITY TESTING

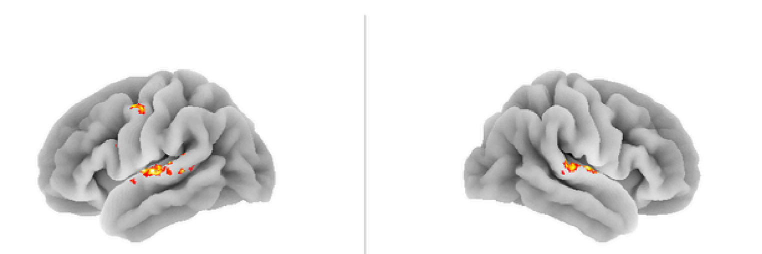
Are punchline events in the right place?

I re-ran the analysis with FWER correction, but with the punchline onsets shifted ±1 TR (1.5 s). If the labels are indeed not arbitrary, we should observe a pattern in these results (which we do!).



Shifted -1.5s

Almost no surviving clusters. Small 'anticipation' signal in the lateral sulcus (some overlap with actual punchline).



Shifted +1.5s

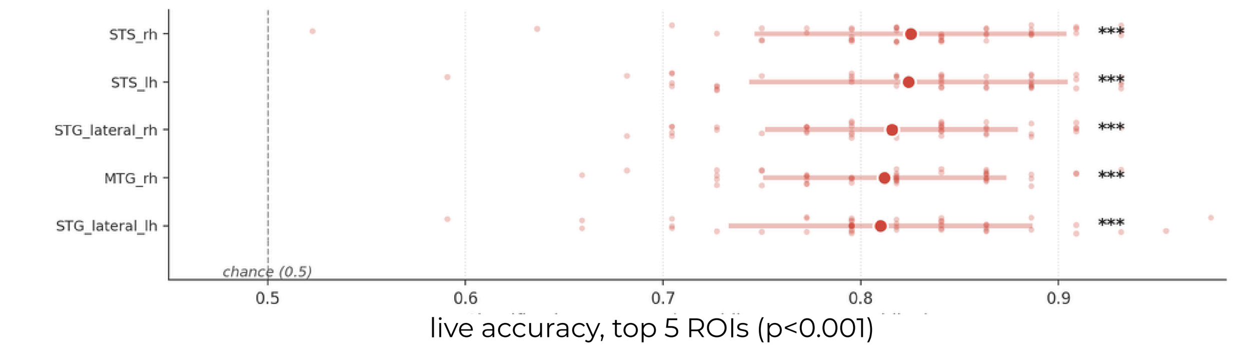
STG clusters shrink, but remain. Motor area cluster gets *bigger* and *brighter*. The delayed motor boost could suggest the live group was physically reacting, laughing along more.

EXPERIMENT II

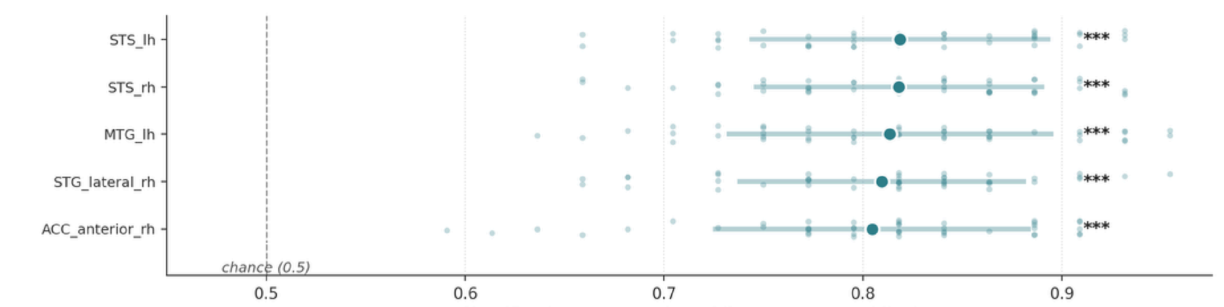
Decoding punchlines from activation patterns via MVPA

Where the GLM treats each vertex independently, multi-variate pattern analysis uses the spatial pattern across a region of interest (ROI). Pipeline:

Randomly sample baseline events ≥12 s from any punchline. Match count to punchlines. → *Estimate per-trial β via least squares separate (LSS).* → *Train linear SVM on trial-by-vertex features and evaluate with leave-one-out cross-validation* → *Group-level two-sample test with FDR correction across ROIs.*



live accuracy, top 5 ROIs (p<0.001)



studio accuracy, top 5 ROIs (p<0.001)

Discussion: all ROIs differentiate punchline and baseline events with significantly above-chance accuracy (70-80%). Most accurate ROIs are consistent with the humor-associated areas identified in first-level GLM in experiment I. However, no significant difference is found in classification accuracy between the live and studio conditions.

LIMITATIONS

Factors to consider

Between-subjects rather than within-subject design

Data is pulled from two completely different studies, each with different subjects on different years using different scanners. This can introduces a lot of noise.

Stories not perfectly identical

Isolated version had jokes cut; delivery differs between live and studio. TR-by-TR methods (e.g., shared response models) weren't viable.

Manual punchline labels

Reviewed, but still subjective. Potential onset errors, others may disagree with definition of punchline (somewhat addressed with staggered onset analysis).

No behavioral / physiological data

No recording of actual laughter, heart rate, or mirth reports.

CONCLUSIONS AND FUTURE WORK

Background laughter amplifies neural response to humor

GLM
Significant Live > Isolated differences in bilateral STG, Heschl's gyri, and left precentral gyrus with FWER correction

MVPA
All tested ROIs decode punchlines reliably over random events, but no significant difference in between-group classification accuracy

Future directions
within-subject design (addressing current study limitations to corroborate results), test functional connectivity at punchline events, look at effect of laugh tracks on *unfunny* content

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